Low-Carbon and Flexible Data Centres



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Today's Lecture

1. **Why data centres matter** — 10 min
   Growth, the UK picture, measuring efficiency
2. **Low-carbon solutions** — 9 min
   Clean and firm supply, heat reuse, cooling
3. **Data centre flexibility** — 12 min
   AI's two-sided role, concepts and trials
4. **Our research** — 22 min
   Quantifying flexibility from a 1 MW data centre

Questions: 5 min

By the end you should be able to:

- Explain why data-centre growth is a power-system challenge
- Compare options for decarbonising data-centre operation
- Explain AI's two-sided role in the net-zero transition
- Define upward and downward flexibility and where it comes from inside a data centre
- Interpret magnitude–duration flexibility results and their limits

01

Why Data Centres Matter

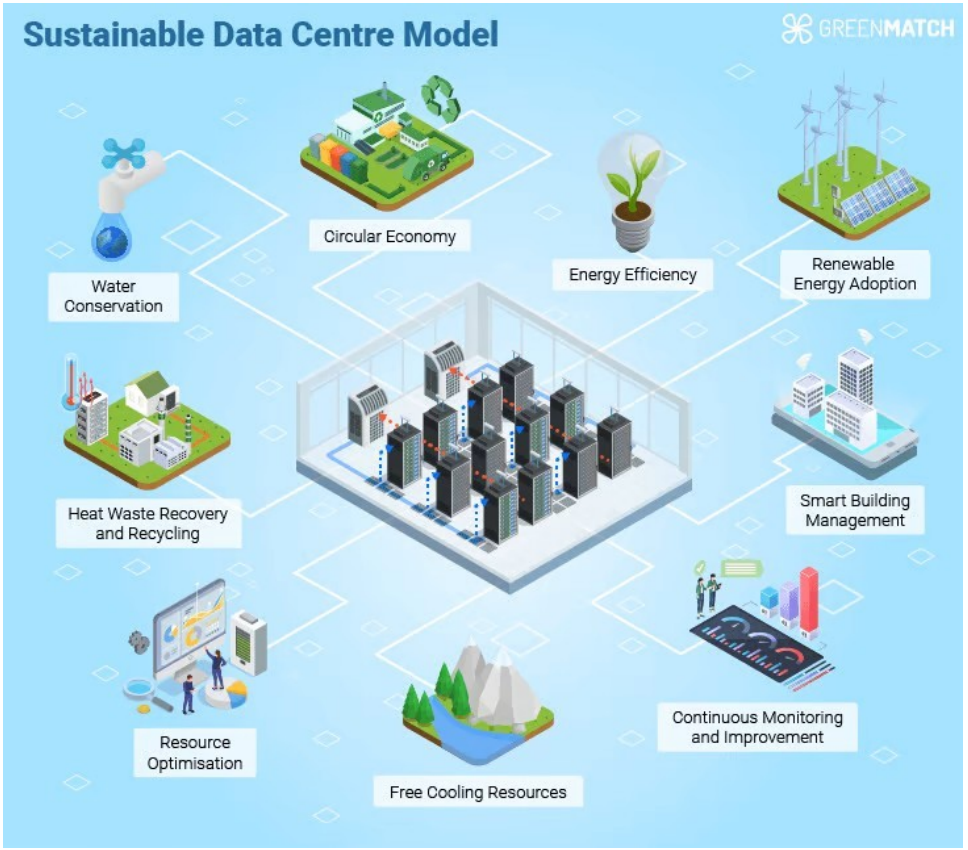
Growth, the UK picture and how we measure efficiency and carbon

What Are Low-Carbon Data Centres?

A low-carbon data centre is designed, built and operated to minimise its environmental impact, above all its carbon footprint.

Five levers

- Efficient energy use (PUE, cooling)
- Clean, time-matched electricity
- Flexibility to support the grid
- Water stewardship and heat reuse
- Smart siting and grid connection



Growth in Data Centres

485 TWh

used by data centres worldwide in 2025: up 17% in one year (global demand +3%)

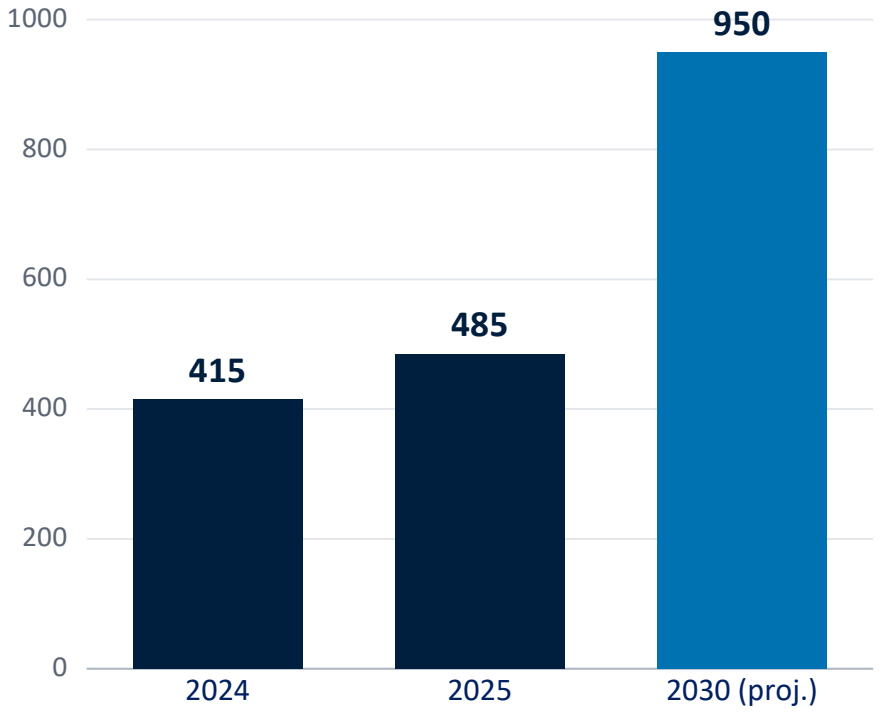
≈950 TWh

projected for 2030: roughly double, ≈3% of global electricity

×3

growth in AI-focused data centres to ≈465 TWh by 2030

Global data-centre electricity use (TWh)

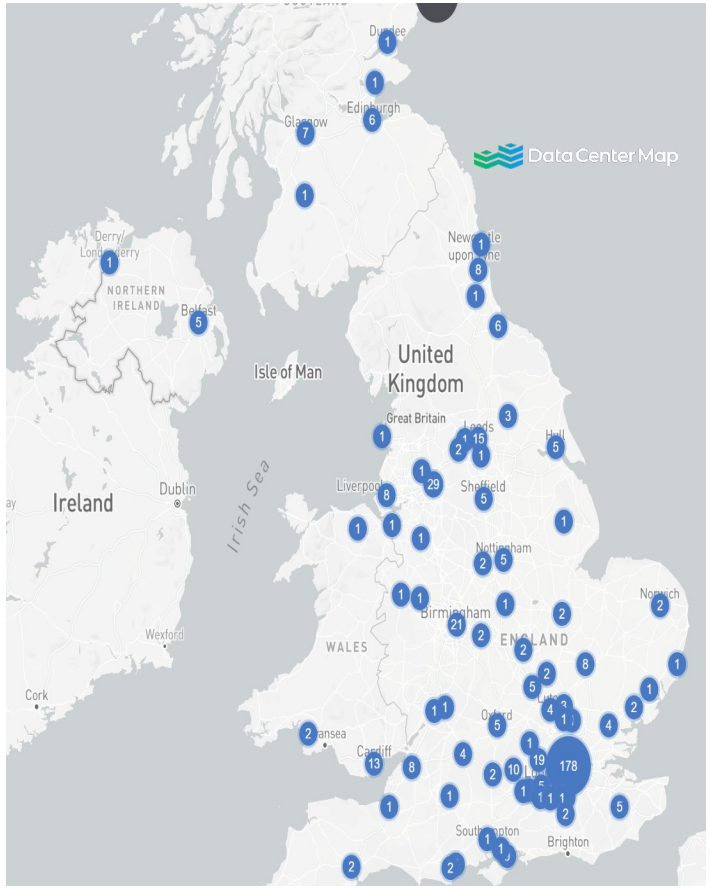


Capital spending by the five largest tech firms: over US$400bn in 2025, +75% expected in 2026

Source: IEA (2026) Key Questions on Energy and AI; IEA (2025) Energy and AI

UK Data Centres

- **Today:** ≈2.4 GW connected, using ≈7.6 TWh a year
- **By 2030:** ≈5.2 GW and over 20 TWh a year expected
- **Connection queue:** over 70 GW of data-centre requests, more than GB's ≈45 GW peak demand
- **AI Growth Zones (5):** Oxfordshire, North East England, North Wales, South Wales and Lanarkshire
- **Wales:** two zones: the M4 corridor (Newport to Bridgend) and Anglesey/Trawsfynydd



Sources: NESO Future Energy Scenarios 2025; NESO evidence to the Environmental Audit Committee (2026); GOV.UK AI Growth Zones (2025–26)

European Data Centres

- Europe's main hubs (Frankfurt, London, Amsterdam, Paris, Dublin: "FLAP-D") are now heavily grid-congested
- Ireland: data centres already use over 20% of national electricity
- **Ireland's new rules (CRU, Dec 2025):** new sites must bring on-site or nearby dispatchable generation or storage, buy 80% renewable electricity, and are assessed on location and flexibility

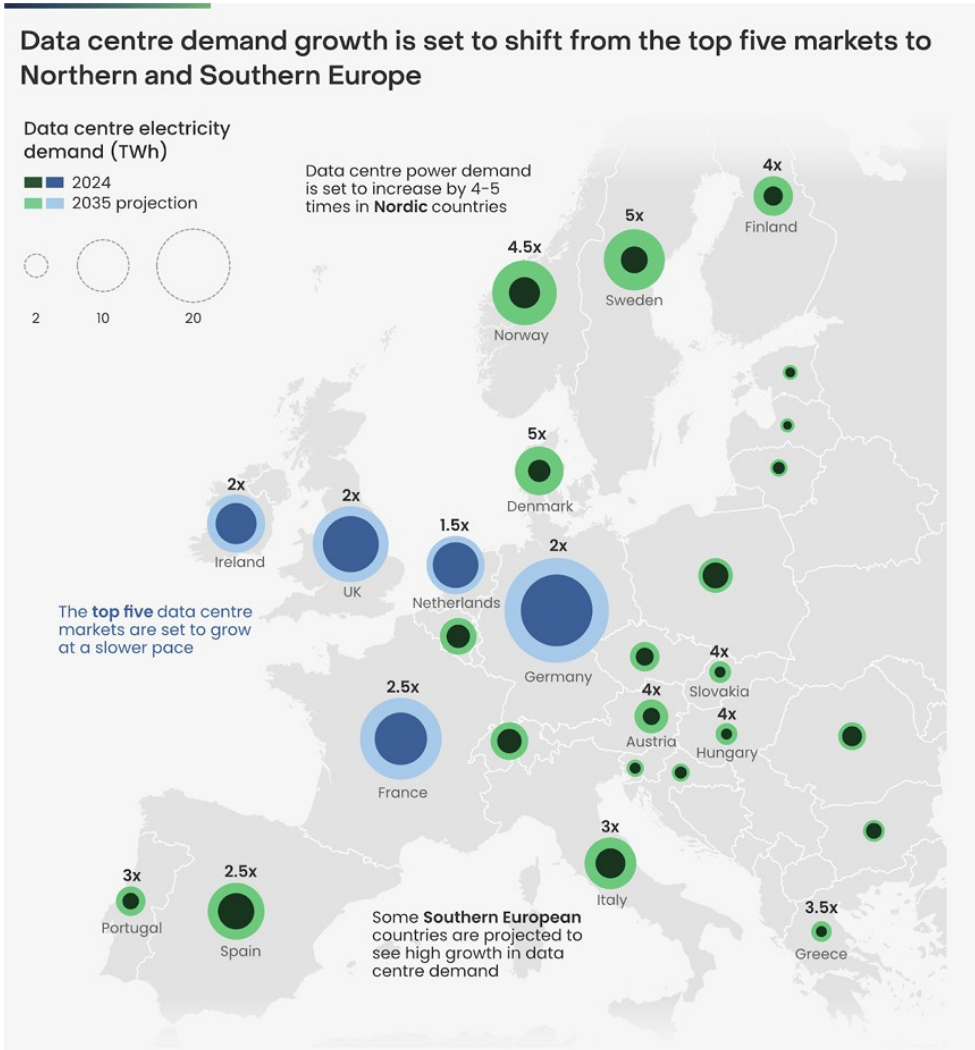
Flexibility is becoming a condition of connection.



[The data center industry beyond FLAP-D. Key national policies and projects making a difference]

Shifting Data Centres Away from Demand

- New builds are moving away from congested demand hubs towards cheap, renewable electricity
- Scotland and Wales: strong wind resource and three AI Growth Zones
- **Trade-off:** cleaner, cheaper power but higher latency: fine for AI training and batch work, not for latency-critical services
- Helps grids and lowers carbon emissions



Measuring Efficiency and Carbon

PUE

Power Usage Effectiveness

Total facility energy ÷ IT energy

1.0 is ideal. Global average ≈1.54, flat for six years; leading hyperscale sites ≈1.1.

WUE

Water Usage Effectiveness

Site water use (L) ÷ IT energy (kWh)

Evaporative cooling saves electricity but consumes water: a trade-off in water-stressed regions.

CUE

Carbon Usage Effectiveness

CO_2 emissions ÷ IT energy

Depends on when and where electricity is used, not only how much.

Annual matching

Buy renewable MWh (PPAs, certificates) equal to annual use. Data-centre renewable PPAs covered over 40 TWh in 2025.

24/7 carbon-free energy

Match every hour with clean supply on the same grid. Much harder: it needs storage, firm low-carbon power and flexibility.

Also count embodied carbon: buildings, servers and batteries (Scope 3).

Sources: Uptime Institute Global Data Center Survey 2025; The Green Grid; IEA (2026) Key Questions on Energy and AI

02

Low-Carbon Solutions

Clean and firm supply, heat reuse and cooling

Solar + Battery Storage

UK: on-site rooftop PV



- Rooftop and car-park PV, often paired with a battery (BESS)
- 500 kW of panels on a Milton Keynes data centre
- The grid stays essential 24/7: on-site PV gives only single-digit % of annual demand without adjacent land

USA: off-grid solar + second-life batteries



- Redwood Energy and Crusoe, Nevada (2025): 12 MW solar + 63 MWh of repurposed EV batteries, built in ≈4 months
- Expanding to ≈20 MW of modular compute (2026)
- Needs sun and cheap land: hard to replicate in northern Europe

Watch: Redwood video

Sources: DCD (2024) On-site rooftop solar at data centers; pv magazine USA (2025, 2026) Redwood/Crusoe second-life battery microgrid

Wind-Powered Data Centres

- **Apatura, Ravenscraig (Motherwell):** 550 MW AI data-centre campus with co-located battery storage on the former steelworks site; £3.9bn, planning in progress
- **Energy angle:** a grid-connected load that can absorb otherwise-curtailed Scottish wind
- **Lanarkshire AI Growth Zone (Jan 2026):** DataVita and CoreWeave, with over 500 MW of on-site power planned
- Scotland is shifting from a small data-centre market to a major build-out region



Apatura plans 500MW data center campus in Scotland

Firm Low-Carbon Power & Backup

Nuclear for 24/7 clean power

- Tech firms are signing nuclear deals: Microsoft–Constellation (Three Mile Island restart), Google–Kairos and Amazon–X-energy (small modular reactors)
- UK: Wylfa, Anglesey chosen for the first Rolls-Royce SMRs (Nov 2025), with power expected from the mid-2030s
- Wylfa sits alongside the North Wales AI Growth Zone

Decarbonising backup power

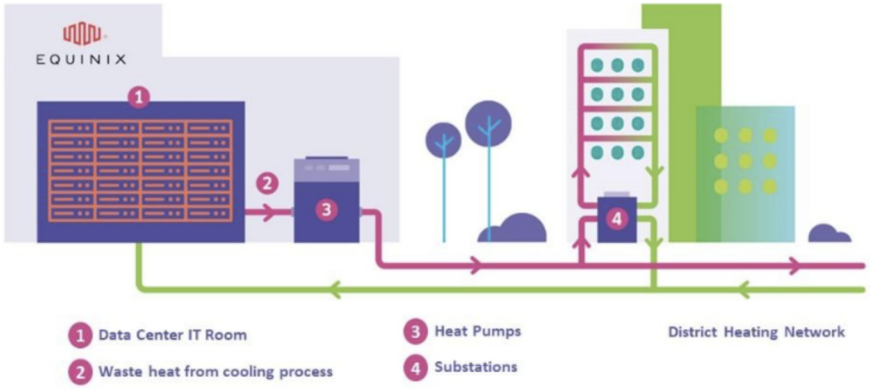
- Diesel generators cover grid outages and run mainly for testing
- Switching to HVO (renewable diesel) cuts lifecycle emissions without new hardware
- Batteries and fuel cells are emerging alternatives, and can also offer grid services

Data centres run 24/7, so wind and solar alone can't match every hour: firm clean power, storage and flexibility fill the gaps.

Sources: GOV.UK (Nov 2025) Wylfa SMR announcement; company announcements (2024)

Reusing Data Centre Heat

- Heat pumps upgrade 20–30 °C waste heat to 50–70 °C for district heating or local use
- Proven at scale in the Nordics; best where heat networks and cold climates exist
- Liquid-cooled AI racks give hotter waste heat, which is easier to reuse



Odense, Denmark (Meta)

≈100,000 MWh of heat a year to the district heating network: enough for ≈7,000 homes. Data centre and heat pumps run on wind power.

Old Oak & Park Royal, London

UK first: £36m from the Green Heat Network Fund to heat ≈10,000 new homes with West London data-centre heat (developer: Hemiko).

Sources: Meta (2020) Denmark data center to warm local community; GLA/OPDC (2023, 2025); DCD (2024) Harnessing waste heat

Cooling in the AI Era

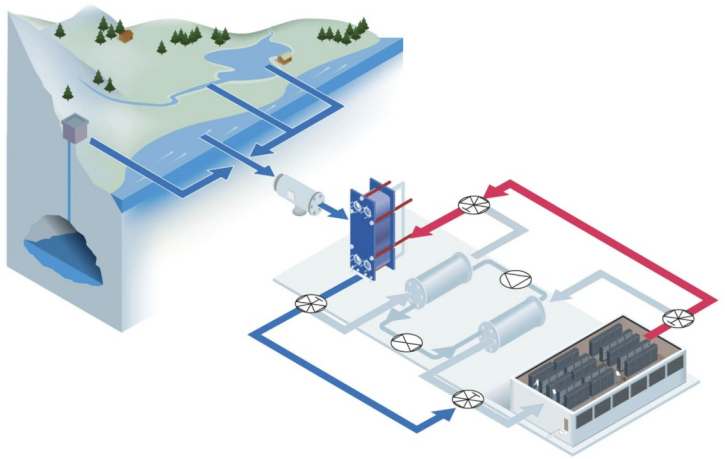
Free cooling

- Use cold outside air or water (sea, river, lake, well) instead of running chillers
- Best in cold climates; manage humidity, temperature and air cleanliness

Liquid cooling

- AI racks draw ≈120–140 kW (e.g. NVIDIA GB200 NVL72); air cooling struggles above ≈30–40 kW per rack
- Direct-to-chip and immersion cooling remove heat at higher temperatures
- Less fan and chiller power, more free-cooling hours, easier heat reuse

Free cooling with a chiller bypass



Cooling and flexibility

Cooling is the largest non-IT load. Thermal mass and chilled-water storage let cooling be shifted in time: a key flexibility source in our research.

Free cooling with water

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03

Data Centre Flexibility

Concepts, real-world trials and what data centres can offer the grid

Flexibility in Power Systems

- **Flexibility:** the ability to change generation or consumption at any time in response to power-system needs
- **Upward flexibility:** increase generation or reduce consumption, which provides power to the system
- **Downward flexibility:** reduce generation or increase consumption, which absorbs surplus power
- Traditionally from fossil-fuel plant; a decarbonised grid needs new sources, including demand

GB flexibility capacity

25.2 GW

in 2024

51–66 GW

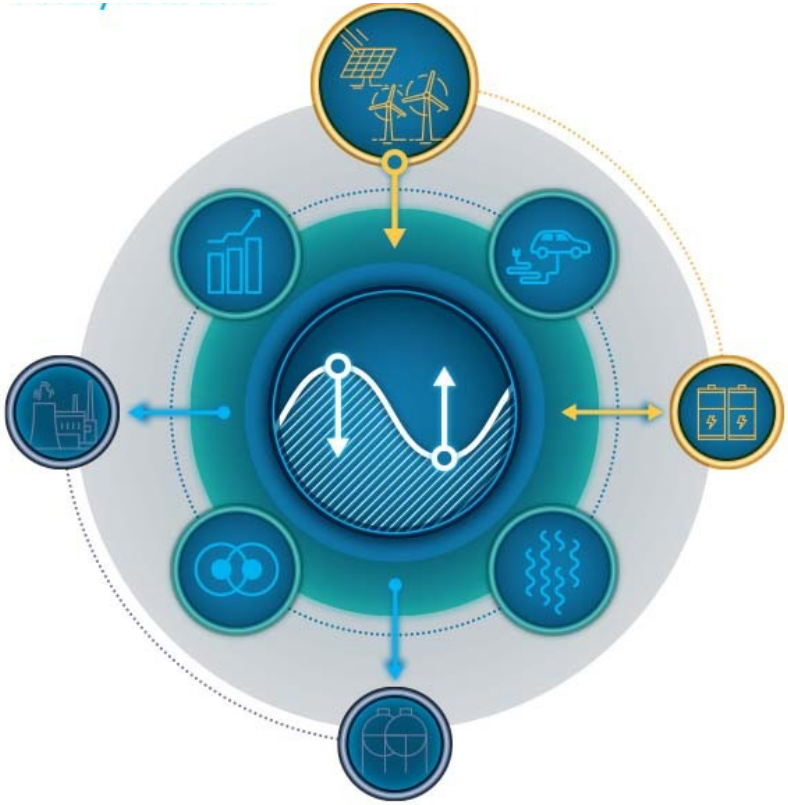
target by 2030

Clean Flexibility Roadmap (DESNZ, Ofgem & NESO, 2025)

Source: DESNZ, Ofgem & NESO (2025) Clean Flexibility Roadmap

Demand-Side Flexibility

- Demand side flexibility (DSF) is the ability of power consumers to modulate their power consumption at any given moment to help support the power system.
- DSF has the potential to be massively impactful (EVs, heat pumps, industrial + commercial loads, etc)
- Data Centres are an increasingly relevant and important candidate for DSF.



Demand-side flexibility to boost power sector transformation

Data Centre Flexibility in Practice

London trial Dec 2025

Emerald AI, National Grid, Nebius, NVIDIA and EPRI: a 96-GPU cluster met 200+ simulated grid events, cutting power by up to 40% while AI jobs stayed within service levels.

Google, USA 2024–26

Demand-response agreements with several US utilities: ≈1 GW of data-centre load that can be curtailed when the grid is stressed.

Phoenix trial 2025

Software-controlled AI cluster cut its power by ≈25% for three hours during a peak-demand event.

Policy 2025–26

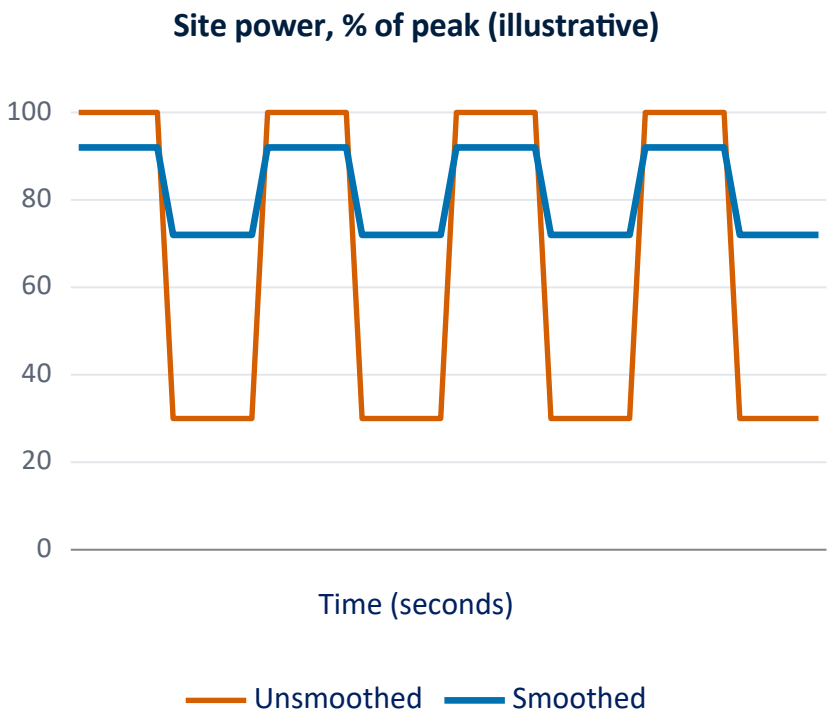
Ireland links connections to on-site generation and flexibility; GB is reforming its connection queue; an industry alliance (Sept 2026) is setting standards.

Industry is already doing this: the open questions are how much, for how long, and from which assets?

Sources: NVIDIA/Emerald AI (2026) UK demonstration; DCD (2026) AI Energy Management Alliance; CRU (2025); Ofgem (2026)

AI Workloads: Flexibility Cuts Both Ways

- Large training jobs synchronise thousands of GPUs: compute and communication phases alternate in lock-step
- Site power can swing by tens to hundreds of MW within seconds, with oscillations around 0.2–3 Hz
- Risk: interaction with generator and network resonances
- Fixes: GPU power smoothing, rack-level energy storage (e.g. NVIDIA GB300), on-site batteries



The same controls that tame these swings can deliver grid flexibility: a large AI site can be a grid problem or a grid resource.

Sources: Choukse et al. (2025) Power Stabilization for AI Training Datacenters; Futurum (2025) NVIDIA GB300 NVL72 power shelf

AI's Two-Sided Role in Net Zero

A driver of emissions

≈350 Mt CO_2

from data centres by 2035: double today, ≈2% of power-sector emissions

- AI runs in data centres that need large, 24/7 electricity supply
- Emissions depend on how that power is generated: over 100 GW of new gas plant is planned to supply data centres directly
- Same 1 TWh: ≈0.4 Mt CO_2 from gas vs ≈0.05 Mt on a low-carbon grid (illustrative)

AI

An enabler of decarbonisation

13 EJ

energy that proven AI uses could save by 2035: ≈3% of global final energy

- Cyber-physical systems that monitor and control emissions in real time
- AI-based modelling and optimisation of energy systems
- Smarter grids: forecast renewables and get more from existing network capacity

The net effect depends on choices: power AI with clean, flexible electricity, and deploy AI where it cuts emissions.

Source: IEA (2026) Key Questions on Energy and AI. Gas ≈400 g/kWh vs low-carbon grid ≈50 g/kWh is illustrative.

How AI Can Speed Up Decarbonisation

Cyber-physical systems: monitor and control emissions

Sense
Smart meters, emissions sensors and IoT devices measure the physical system

Analyse
AI detects anomalies and predicts demand, emissions and failures

Act
Controllers adjust HVAC, industrial processes and power dispatch

A closed loop: e.g. building energy management, industrial process control, real-time grid monitoring

AI-based modelling and optimisation of energy systems

Forecasting
Wind, solar, demand and prices

Optimisation
Dispatch, storage and flexibility scheduling

Planning
Grid expansion and digital twins of networks

Maintenance
Predict failures in transformers and turbines

Not a silver bullet: digital skills, fragmented data and cybersecurity limit uptake. Our research (section 4) is one example: optimisation that makes a data centre a flexible grid asset.

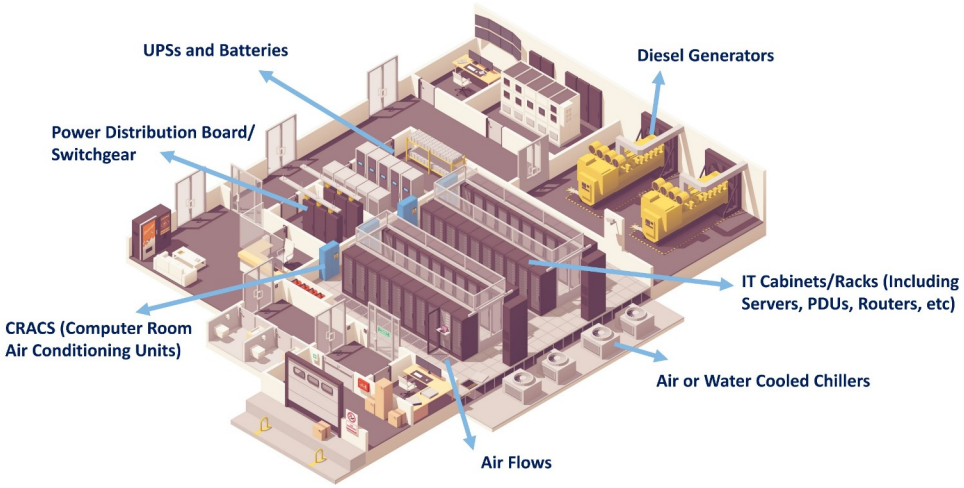
Source: IEA (2026) Key Questions on Energy and AI; IEA (2025) Energy and AI

Flexibility from Data Centres

- **IT Workload:** IT workloads, less sensitive to delay, are re-scheduled to adjust processing power and its associated electricity consumption of the DC.

- **UPS Battery:** The UPS battery is a key DC asset whose core responsibility is providing back up power to the IT servers, in case of a grid outage, before the diesel generators have powered up.

- **Cooling System:** This comprises chiller, CRAC/HVAC units and TES tank. The TES tank, as well as the thermal mass of the components in the DC, are used to provide flexibility in the cooling system.



04

Our Research

Characterisation and Quantification of Data Centre Flexibility for Power System Support

J. Day, M. T. Takci, M. Qadrdan · Submitted to *Applied Energy* (2026)

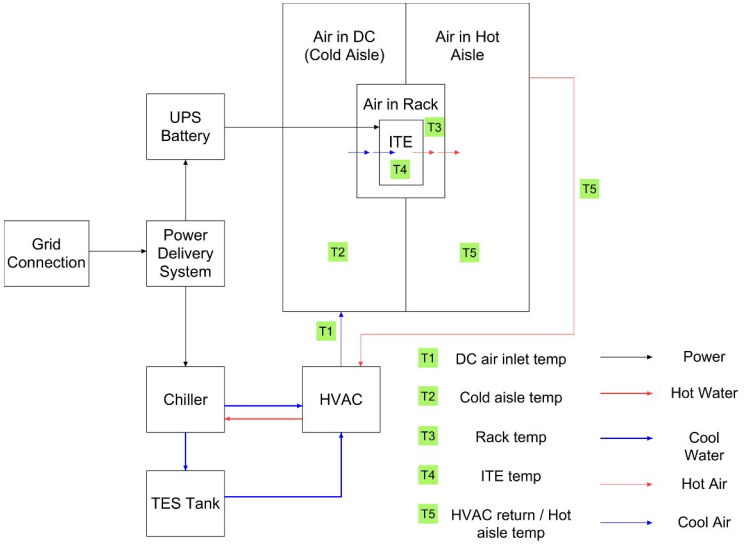
Research Question

Research question

For a requested start time and power change, how long can a data centre sustain it while respecting IT service, storage, thermal and recovery constraints?

Contributions

1. Integrated model: IT workload, UPS battery, cooling and thermal storage co-optimised
2. Duration-aware flexibility assessment, including recovery after each event
3. Full-year (2025) evaluation of electricity cost savings and sensitivities



Data-centre layout: grid connection, UPS, chiller, TES tank, hot and cold aisles

Case study

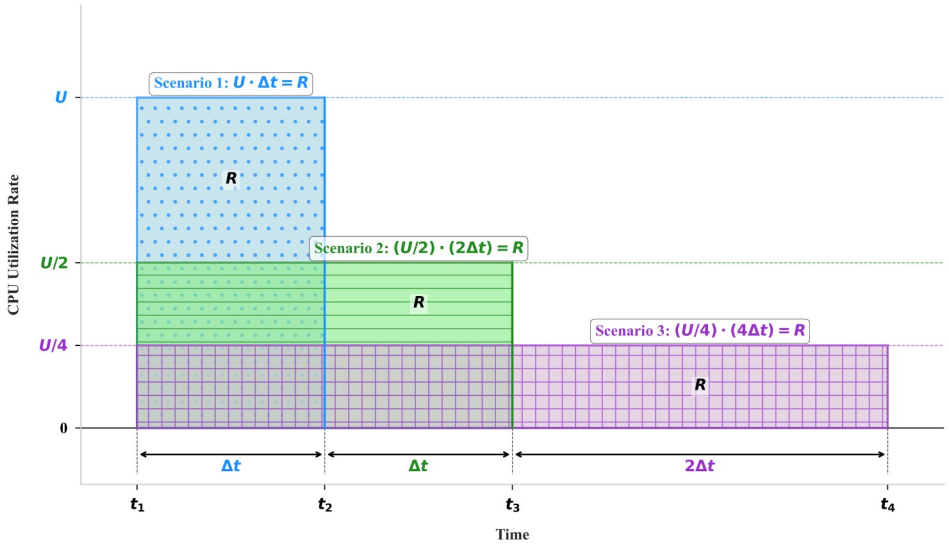
Hypothetical 1 MW (IT) data centre
GB 2025 day-ahead prices
15-minute resolution, full year

Modelling IT Workload Flexibility

- Each job needs a fixed amount of compute: **R = CPU utilisation × time**
- Run it slower for longer or faster for shorter: the work done is the same
- Server power rises with utilisation:

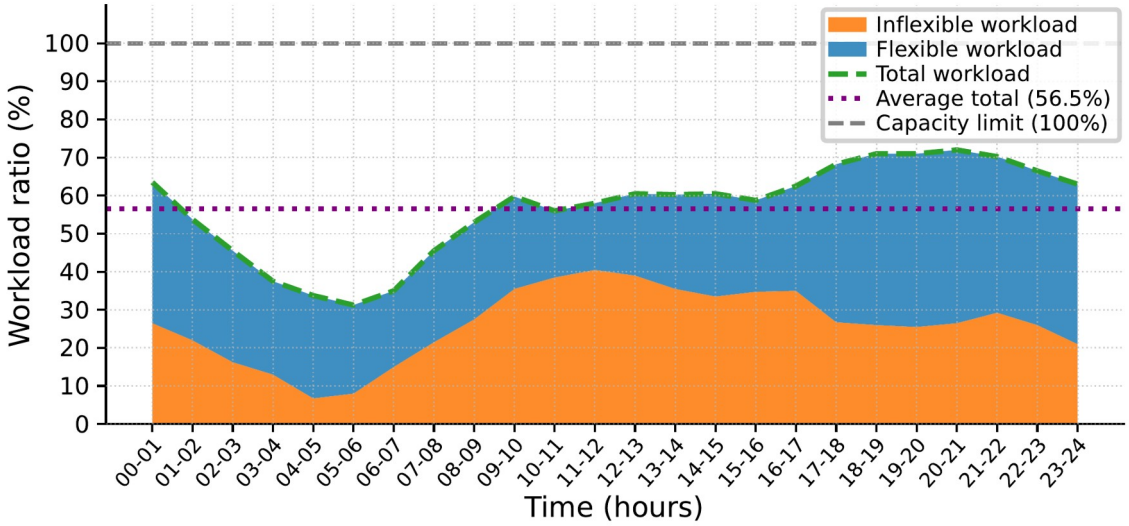
$$P_{IT} = P_{idle} + (P_{max} - P_{idle}) \cdot u^{1.32}$$

- Every job must finish before its deadline (deferral window)
- All servers are represented as one aggregate server



Day, Takci & Qadrdan (2026), Fig. 3 and Eq. (1)

Workload Assumptions



- Google trace: up to ≈70% of work is delay-tolerant
- Alibaba trace: flexible jobs are 40% of jobs but 70% of energy
- Adopted average utilisation: 56.5%
- Inputs are literature-based assumptions, not one site's measurements

Tranche	Latest start after arrival	Share of flexible work (hourly range)
k = 1	30 min	10–50%
k = 2	60 min	12–28%
k = 3	2 h	13–30%
k = 4	3 h	15–47%

Day, Takci & Qadrdan (2026), Fig. 2 and Tables III–IV

Power and Cooling Assets

UPS battery

600 kWh

- Half kept in reserve for backup
- Charges at up to 270 kW
- Efficiency 0.82 charge / 0.92 discharge
- Discharge supplies IT load, cutting grid import

Chiller + thermal mass

COP 5

- Chiller up to 400 kW electrical
- Cold aisle kept at 18–23 °C
- Air, racks and servers store heat: a short buffer
- Less IT work means less heat and less cooling

Thermal energy storage

1,000 kWh

- Stores cooling as chilled water
- Charges/discharges at up to 300 kW thermal
- Efficiency 0.9 each way
- Lets the chiller run when power is cheap

Coupled assets: deferring IT work cuts heat and cooling; UPS discharge supports IT; cooling is limited by temperatures and TES charge. A change in one shifts what the others can do.

Method: Three Scenarios

Scenario 1

Benchmark

Work runs as it arrives; UPS and TES idle; cooling still cost-optimised within temperature limits

Output: Benchmark annual cost

Scenario 2

Cost-optimised year

Co-optimise IT, UPS, cooling and TES for all 365 days of 2025: daily optimisations with a 3 h look-ahead

Output: Optimised schedule, annual saving and baseline grid power

Scenario 3

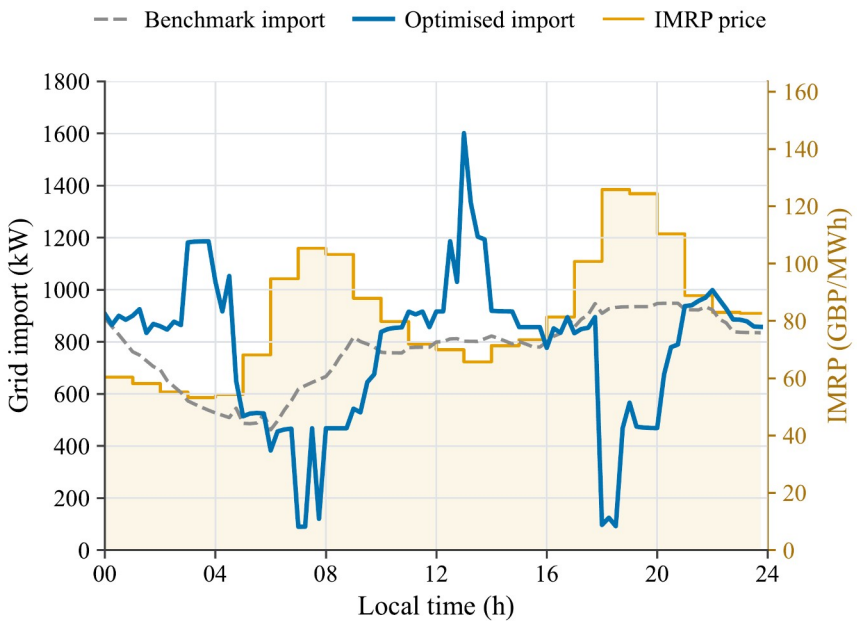
Flexibility duration

From the optimised baseline, find how long each requested power change can be held, including recovery

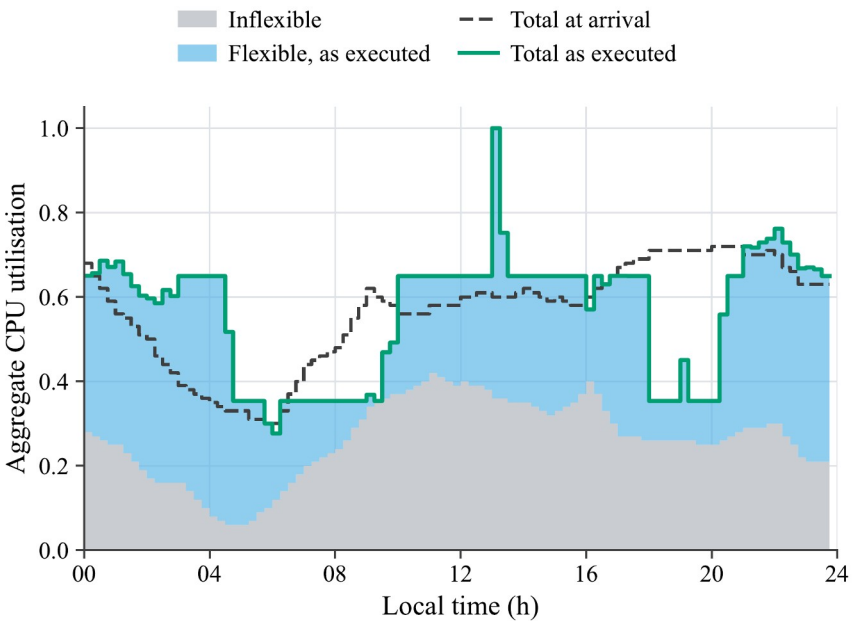
Output: Magnitude–duration heatmap and asset contributions

Setup GB 2025 day-ahead reference prices (incl. 672 negative-price quarter-hours) · 15-minute resolution: 35,040 intervals · mixed-integer linear programming in Pyomo, solved with HiGHS · cold aisle 18–23 °C in every scenario

A Day of Optimised Operation: 22 Sept 2025



(a) Grid import and price

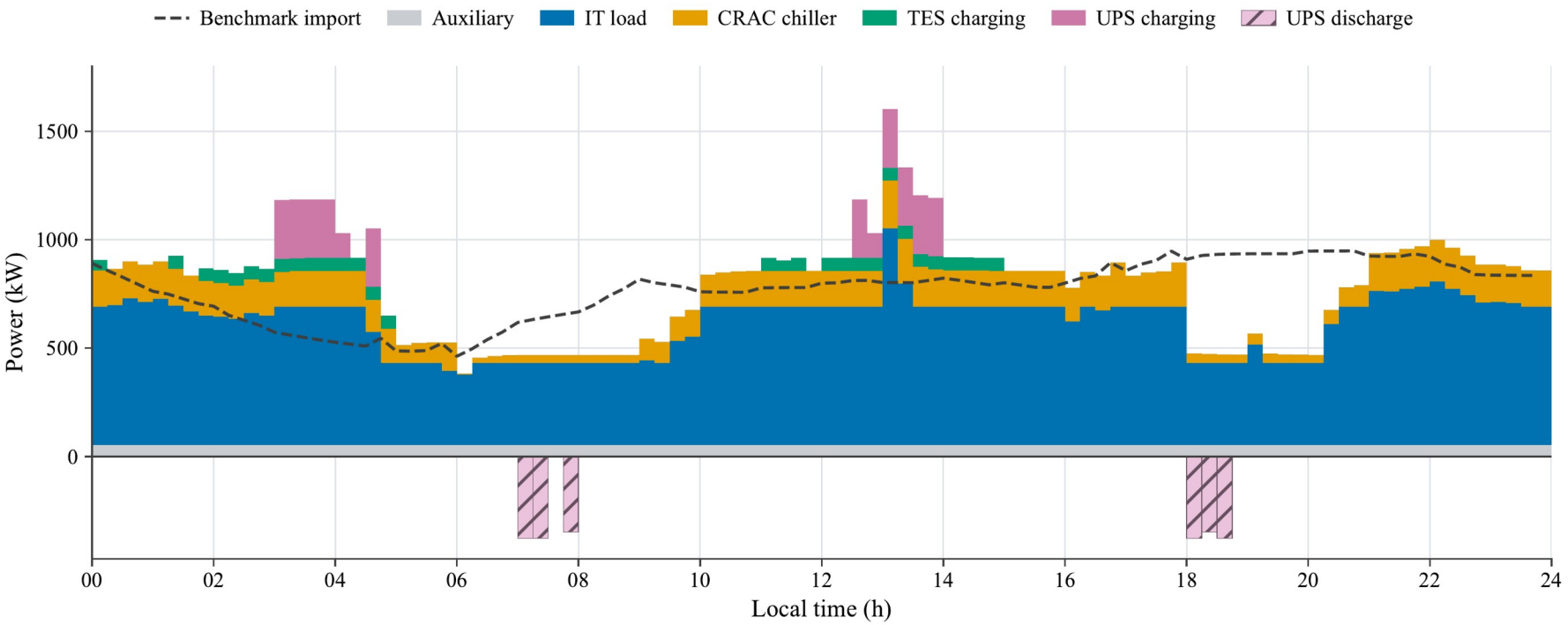


(b) Workload: arrivals vs execution

- Import drops at the morning and evening price peaks and rises when power is cheap
- Flexible work (blue) is deferred within its window, never past its deadline
- 22 Sept was the most typical day of 2025; the annual saving comes from all 365 days

Day, Takci & Qadrdan (2026), Fig. 5

Optimised Dispatch by Component



Charge when cheap

TES and UPS charge in low-price periods, sitting above the IT load

Discharge at peaks

UPS discharge (hatched) supplies IT load at price peaks, cutting grid import

Cooling follows IT

Deferring work lowers heat, so chiller (orange) demand falls too

Day, Takci & Qadrdan (2026), Fig. 6

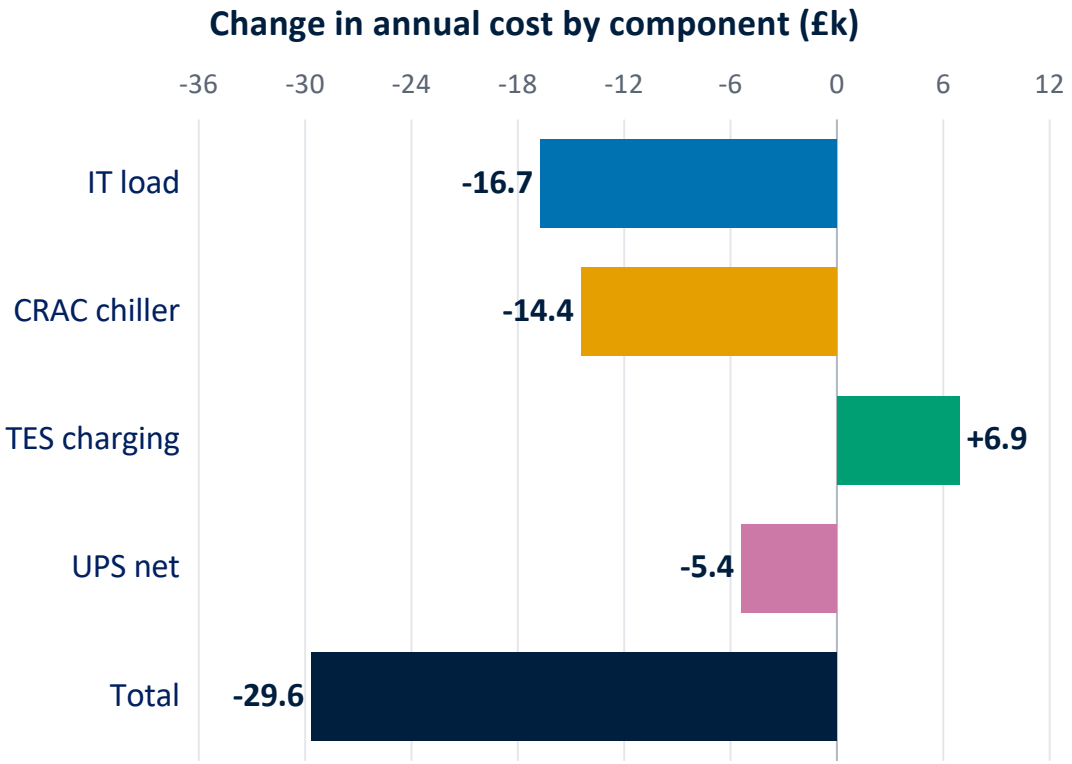
Annual Results: 5.4% Lower Electricity Cost

£29,627

saved in 2025 (£550,504 → £520,877)

5.38%

lower annual electricity cost



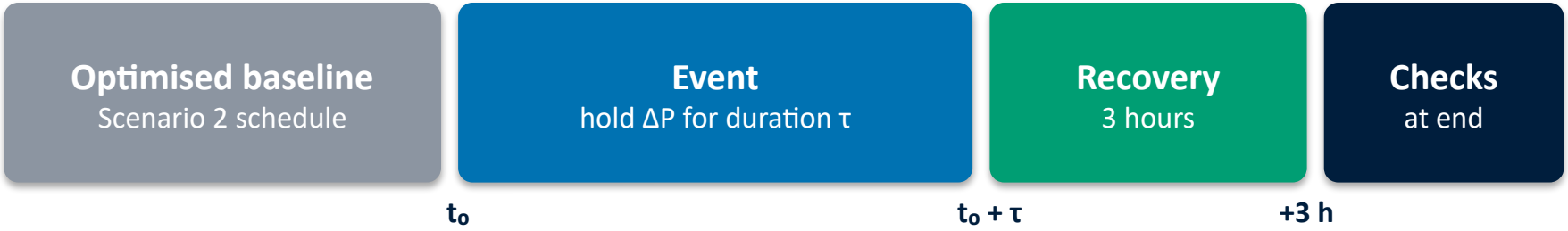
But cheaper ≠ lower energy or peak

- Grid energy rises 1.05% (6.718 → 6.789 GWh)
- Peak import rises from 975 kW to 1,683 kW
- Price signals alone can increase network peaks
- Excludes battery degradation, network tariffs and flexibility revenue

Discussion: what would a carbon- or network-aware objective change?

Day, Takci & Qadrdan (2026), Section 5.3 and Table VI

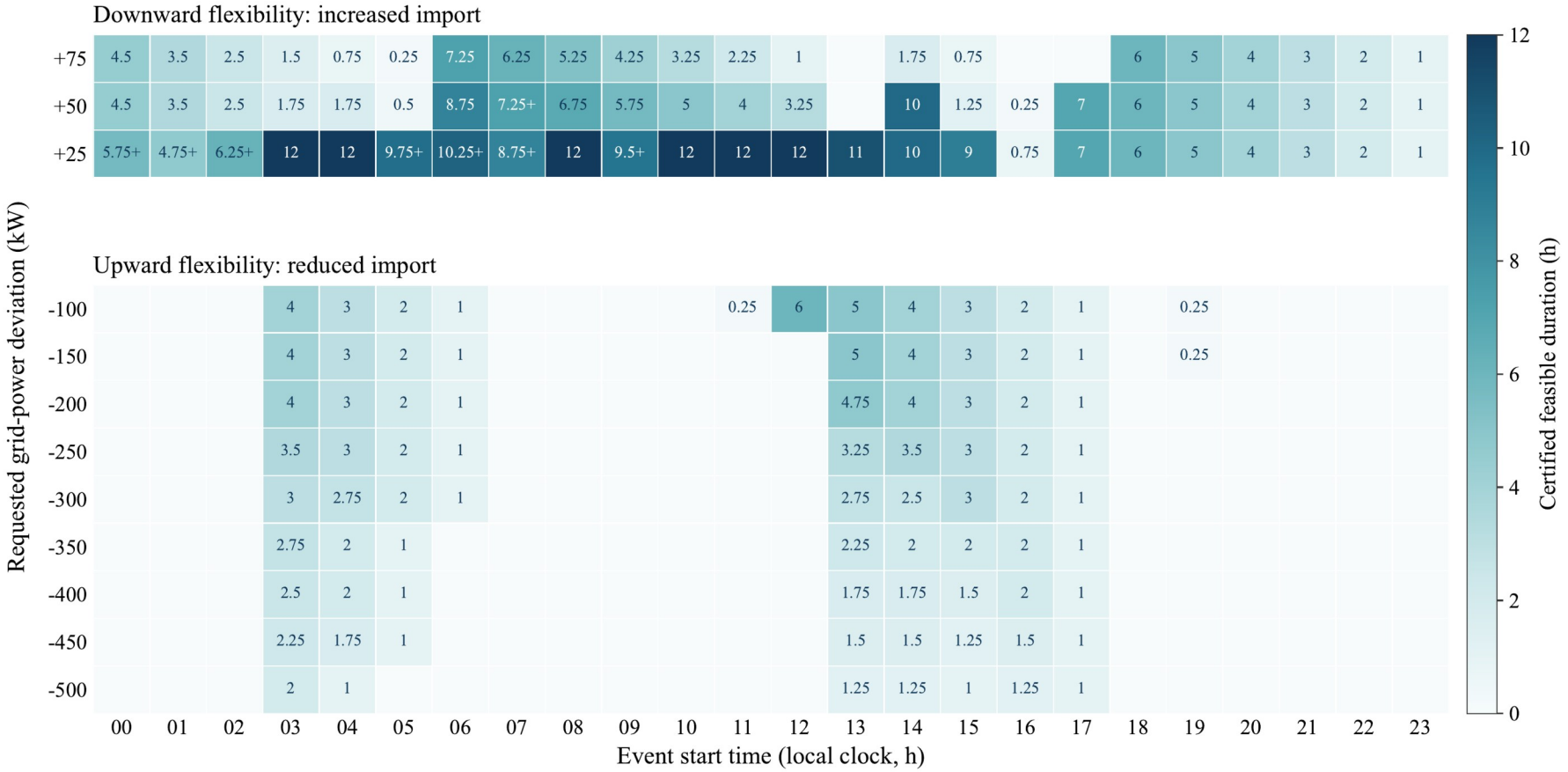
Scenario 3: How Long Can Flexibility Last?



End checks: storage ≥ baseline · temperatures within 0.05 °C · no extra unfinished work

- **288 requests on 22 Sept:** 24 hourly start times × 12 power changes
- **Upward** (reduce import): −100 to −500 kW in 50 kW steps
- **Downward** (increase import): +25, +50 and +75 kW
- Duration searched in 15-minute steps, capped at 12 h or the end of the day
- **Certified feasible duration:** proven possible, not necessarily the maximum
- No flexibility-service revenue is counted

Magnitude vs Duration of Flexibility

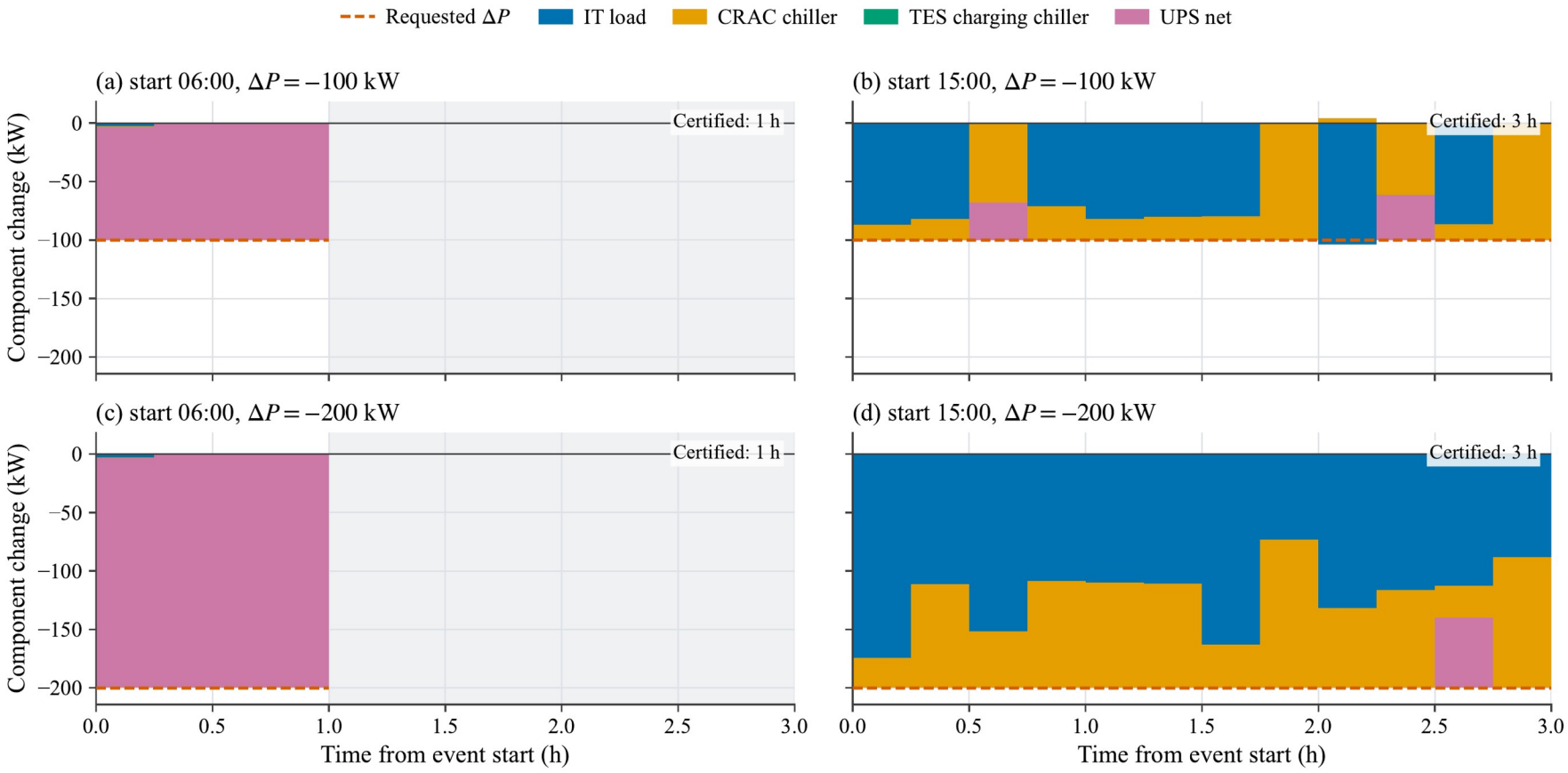


–100 kW lasts 1 h from 06:00 but 6 h from noon

Larger requests generally last for shorter times

+25 kW reaches the 12 h cap at six start times

Upward Flexibility: Who Delivers It?

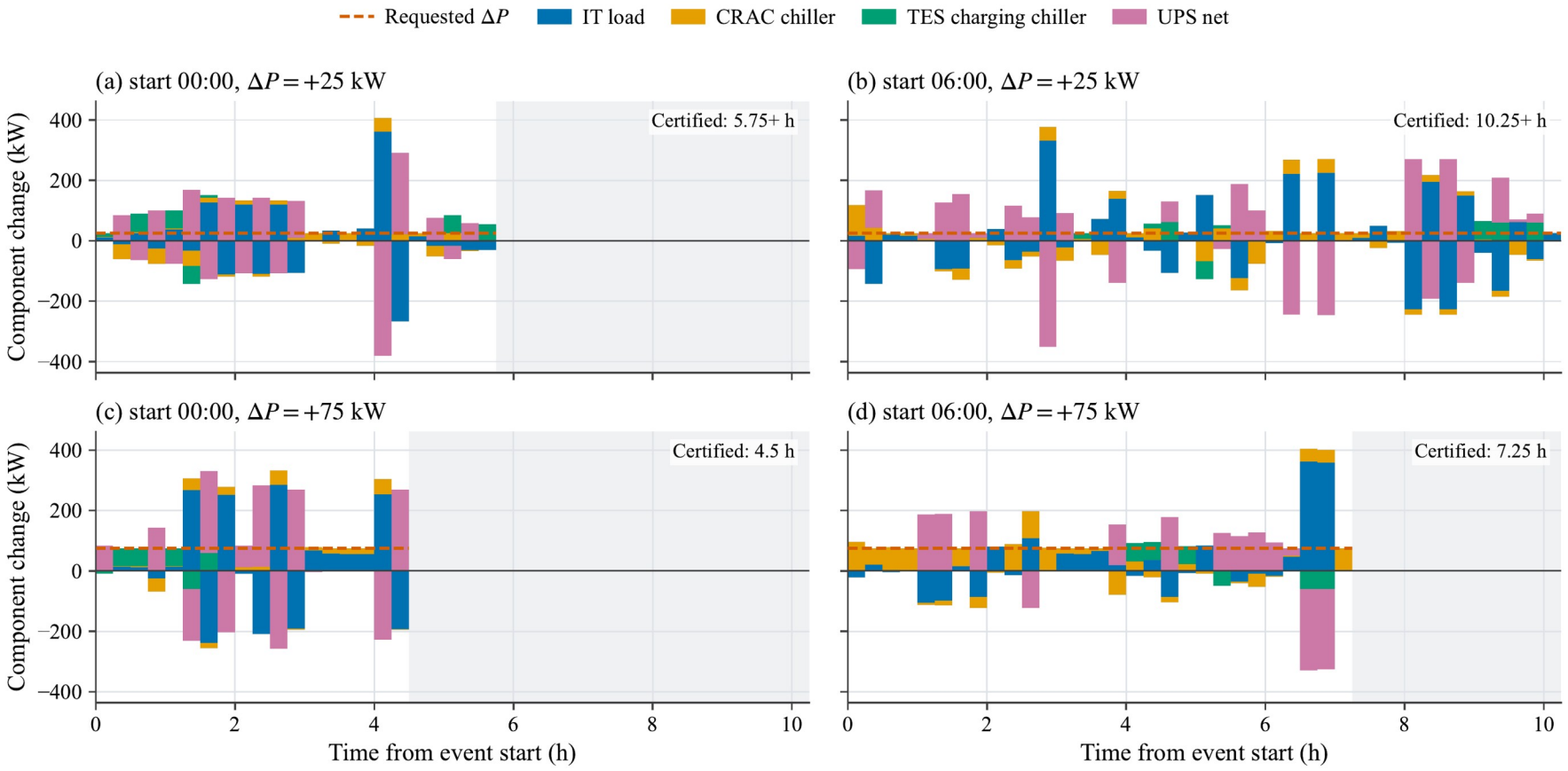


06:00 start: the UPS battery delivers almost all of it, for 1 h

15:00 start: IT deferral plus lower cooling, for 3 h

Operating state, not installed capacity, decides who responds

Downward Flexibility: Assets Work Together

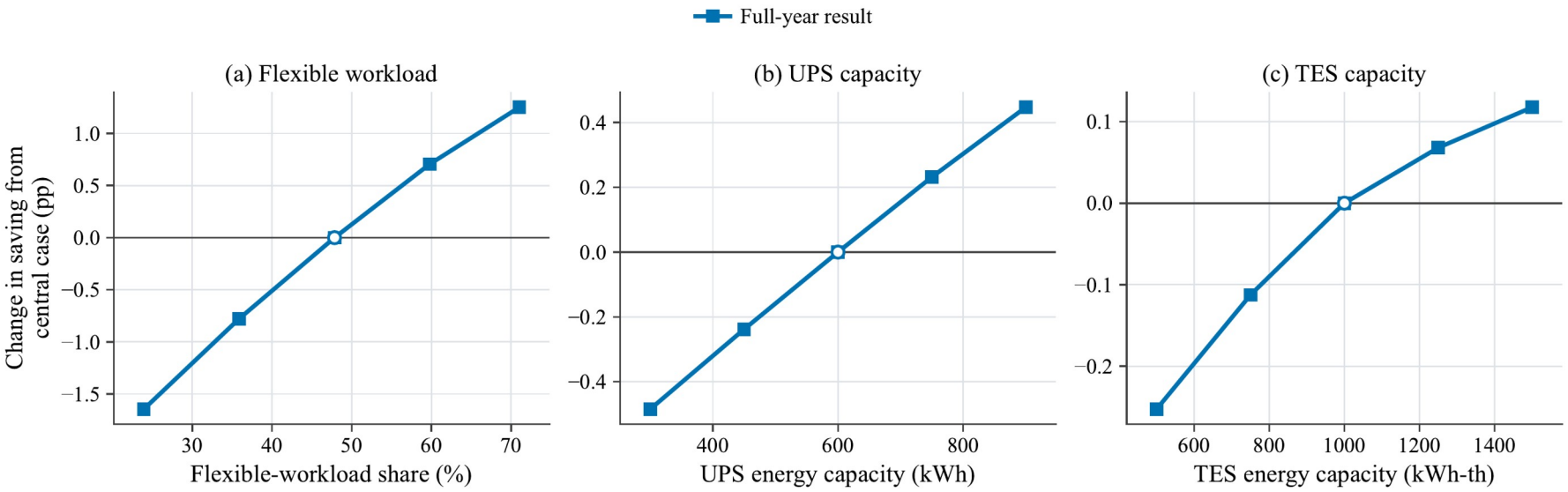


Assets often move in opposite directions: one recharges while another over-delivers

From 06:00: +25 kW for ≥10.25 h; +75 kW for 7.25 h

Only work that has already arrived can be run early

Sensitivity: What Drives the Savings?



Flexible workload share

3.7% → 6.6%

annual saving at 0.5× → 1.5×

UPS energy capacity

4.9% → 5.8%

annual saving at 0.5× → 1.5×

TES energy capacity

5.1% → 5.5%

annual saving at 0.5× → 1.5×

Bigger UPS? +150 / +300 kWh saves £1,275 / £2,460 a year, worth up to ≈£55–57 per added kWh (8% real rate, 10 years). An expenditure threshold, not a business case.

Why Model the Whole Facility?

1

IT and cooling are coupled

Deferring work also cuts cooling, which an IT-only or cooling-only model would miss.

2

Timing decides the asset

Morning reductions use the UPS; afternoon reductions use IT deferral and cooling.

3

Assets buffer each other

TES, UPS and IT can move in opposite directions so the net response holds.

Offer flexibility as magnitude × duration × start time, not a single kW number.

Our outputs are candidate offers: network limits and market rules are the next step.

Ongoing and Future Work

Revenue stacking

Value flexibility across balancing, capacity and local network markets

Pre-conditioning

Prepare before downward events so more IT work can take part

Portfolios

Aggregate many sites and shift work between locations

Network impact

Test offers against network limits, hosting capacity and peaks

AI and real data

GPU power management and real operator workload data

Seasons and design

Seasonal ambient/COP effects; size storage for flexibility at the design stage

Key Takeaways

1. Data-centre demand is growing fast, and grid connections are now the bottleneck
2. Low-carbon operation needs efficiency, clean and firm supply, heat reuse and flexibility together
3. Flexibility depends on timing: in our 1 MW case, 100 kW lasts 1 h at 06:00 but 6 h at noon
4. Cheaper is not automatically greener: objectives and price signals matter

Questions?

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Paper: Day, Takci & Qadrdan, submitted to *Applied Energy* (2026)

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